Saksham Upreti

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EXPERIENCE

BIOWELLTH — Software Engineering Intern Jan 2026 – Present

- Integrated **Microsoft Azure AI Foundry** capabilities into the production application, evaluating new AI features and wiring them into the existing product
- Diagnosed and fixed frontend bugs and shipped UI improvements, improving end-user experience and overall product stability
- Delivered features across the **full stack**, building backend functionality alongside frontend work

PROJECTS

Distributed RAG System | Python, FastAPI, gRPC, FAISS, Redis, PostgreSQL, Docker Aug 2026 – Present

- Architected a horizontally-sharded retrieval-augmented generation pipeline, implementing **consistent hashing from scratch** to distribute vector embeddings across multiple FAISS index nodes with minimal data movement on node join/leave
- Built a **scatter-gather query coordinator over gRPC** that queries all shards concurrently, merges ranked results, and tolerates partial shard failures via timeouts
- Added shard replication, health-check-based failover, Redis caching and end-to-end **OpenTelemetry** tracing for reliability and observability

DocuMind | Next.js, TypeScript, PostgreSQL/pgvector, Voyage AI, Anthropic API Aug 2026 – Present

- Built a full-stack AI document-chat app: users upload PDFs and hold context-aware Q&A over their content through a complete **RAG pipeline** (chunking, Voyage AI embeddings, pgvector retrieval, grounded generation with the Anthropic API)
- Implemented authentication with Clerk, file storage with Supabase Storage, and a Next.js (App Router) frontend with a custom Tailwind CSS design system
- Implementing **Contextual Retrieval with prompt caching** to improve retrieval accuracy beyond standard chunk-based RAG

Argus — AI Code Review | Next.js, TypeScript, pgvector, Anthropic API In development

- Building a **GitHub App** that reviews every pull request with **full-repository context** via retrieval over a vector index, instead of diff-only analysis, to catch duplicated logic and conflicts with existing patterns
- Designed the pipeline: GitHub webhooks → Next.js route handlers → pgvector retrieval (Voyage AI) → Claude-generated structured findings posted back as inline PR comments
- Completed GitHub App + OAuth (Auth.js) setup and database foundations; webhook and diff pipeline in progress, with a Semgrep security pass and a quality dashboard planned

EDUCATION

Bipin Tripathi Kumaon Institute of Technology — B.Tech, Computer Science 2023 – 2027 (Expected)

Coursework: Data Structures, Algorithms, Databases, Operating Systems, Machine Learning

SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Web & Backend: Next.js (App Router), React, Tailwind CSS, FastAPI, REST APIs, gRPC, GitHub Apps / Octokit, Auth.js, Clerk

AI / RAG: Retrieval-Augmented Generation, Vector Embeddings, pgvector, FAISS, Contextual Retrieval, Prompt Engineering, Anthropic API, Azure AI Foundry

Databases & Infra: PostgreSQL, Supabase, Redis, Docker, OpenTelemetry, Semgrep

Mobile: Flutter, Dart, Firebase (Auth/Firestore/FCM)

Tools: Git/GitHub, Claude Code, Cursor